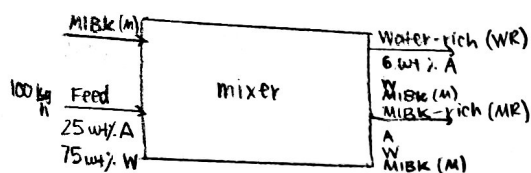


1.



From Figure 5.15.

Water-rich phase $\rightarrow$ tie-line $\rightarrow$ MIBk-rich phase

6 wt% A
2 wt% MIBk
92 wt% W

5 wt% A
92 wt% MIBk
3 wt% W

Material Balance: A

$$0.25 \cdot 100 = 0.06WR + 0.05MR$$

$$25 = 0.06WR + 0.05MR$$

Material Balance: M

$$1.0 \cdot M = 0.02WR + 0.92MR$$

$$M = 0.02WR + 0.92MR$$

Material Balance: W

$$0.75 \cdot 100 = 0.92WR + 0.03MR$$

$$75 = 0.92WR + 0.03MR$$

$$\begin{bmatrix} 0.06 & 0.05 \\ 0.92 & 0.03 \end{bmatrix} \begin{bmatrix} WR \\ MR \end{bmatrix} = \begin{bmatrix} 25 \\ 75 \end{bmatrix}$$

$$WR = 67.87 \text{ kg/h}$$

$$MR = 418.55 \text{ kg/h}$$

$$M = 0.02[67.87] + 0.92[418.55]$$

$$M = 386.42 \text{ kg/h}$$

$$\frac{\text{kg solvent}}{\text{kg feed}} = \frac{M}{F} = \frac{386.42}{100} = 3.86$$

✓

$$\frac{25}{25}$$

1) 25

2) 25

3) 20

4) 25

$$\frac{95}{100}$$

2.

Phenol: C_6H_5OH

$$6(12.01) + 6(1.008) + 1(15.999) = 94.107 \text{ g/mol}$$

Conversion

$$1 \text{ g Phenol} \cdot \frac{1 \text{ mol Phenol}}{94.107 \text{ g Phenol}} \cdot \frac{1000 \text{ mmol}}{1 \text{ mol}} = 10.63 \text{ mmol}$$

Phenol concentration $\cdot$ Volume + Phenol adsorbed $\cdot$ Mass = 10.63 mmol

$$P_c \cdot V + P_a \cdot M = 10.63 \text{ mmol}$$

$$0.6 \frac{\text{mmol}}{\text{L}} \cdot 1 \text{ L} + 2.0 \frac{\text{mmol}}{\text{g}} \cdot 5 \text{ g} = 10.6 \approx 10.63$$

$$\frac{2.0 \frac{\text{mmol}}{\text{g}} \cdot 5 \text{ g}}{10.63 \text{ mmol}} = 94\%$$

$$\frac{25}{25}$$

1) Air

$$T_0 = 25^\circ\text{C} \quad T_1 = 150^\circ\text{C}$$

$$\Delta \hat{H} = 3640 \text{ J/mol}$$

$$\text{Flow rate} = 1.25 \text{ m}^3/\text{min}$$

$$P = 122 \text{ kPa}$$

Conversion

$$122 \text{ kPa} \times \frac{1000 \text{ Pa}}{1 \text{ kPa}} = 122,000 \text{ Pa}$$

$$T(\text{K}) = 150 + 273.15 = 423.15 \text{ K}$$

$$PV = nRT \rightarrow n = \frac{PV}{RT}$$

$$P = 122,000 \text{ Pa}$$

$$\dot{V} = 1.25 \text{ m}^3/\text{min}$$

$$R = 8.314 \frac{\text{Pa} \cdot \text{m}^3}{\text{K} \cdot \text{mol}}$$

$$T = 423.15 \text{ K}$$

$$\dot{n} = \frac{122,000 \text{ Pa} \cdot \frac{1.25 \text{ m}^3}{\text{min}}}{8.314 \frac{\text{Pa} \cdot \text{m}^3}{\text{K} \cdot \text{mol}} \cdot 423.15 \text{ K}}$$

$$\dot{n} = 43.35 \frac{\text{mol}}{\text{min}} \cdot \frac{1 \text{ min}}{60 \text{ s}}$$

$$\dot{n} = 0.72 \frac{\text{mol}}{\text{s}}$$

$$\Delta \hat{K}E^0 + \Delta \hat{P}E^0 + \Delta \hat{H} = Q + \hat{W}_s^0$$

$$\dot{Q} = \dot{n} \Delta H$$

$$Q = 0.72 \frac{\text{mol}}{\text{s}} \cdot \frac{3640 \text{ J}}{\text{mol}}$$

$$Q = 2620.8 \text{ W} \times \frac{1 \text{ kW}}{1000 \text{ W}}$$

$$Q = 2.62 \text{ kW}$$

20
25

b) ΔE_k would be positive since there is a temperature increase but it would be small (almost negligible). There are several formulas: $E_k = \frac{1}{2}mv^2$ and $E_k = \frac{3}{2} \frac{R}{N_A} T$

$$E_k = \frac{3}{2} \frac{8.314}{6.022 \times 10^{23}} \cdot 423 = 8.76 \times 10^{-21}$$

more information is needed

4. 1 g mol H_2O

a) 0 km/h to 100 km/h

$$\Delta E_k = \frac{1}{2} m \Delta v^2$$

$$\Delta E_k = \frac{1}{2} \left[1 \text{ g mol} \cdot \frac{18 \text{ g}}{\text{g mol}} \right] \left[\left(100 \frac{\text{km}}{\text{h}} \right)^2 - \left(0 \frac{\text{km}}{\text{h}} \right)^2 \right]$$

$$\Delta E_k = 90000 \frac{\text{g km}^2}{\text{h}^2} \left(\frac{1 \text{ kg}}{1000 \text{ g}} \cdot \left(\frac{1 \times 10^3 \text{ m}}{1 \text{ km}} \right)^2 \cdot \left(\frac{1 \text{ h}}{3600 \text{ s}} \right)^2 \cdot \frac{\text{J}}{\text{kg} \cdot \text{m}^2/\text{s}^2} \right) = 6.94 \text{ J}$$

b) rising 100 m

$$\Delta E_p = m g \Delta h$$

$$\Delta E_p = \left[1 \text{ g mol} \cdot \frac{18 \text{ g}}{\text{g mol}} \right] \cdot \left[9.81 \frac{\text{m}}{\text{s}^2} \right] [100 \text{ m} - 0 \text{ m}]$$

$$\Delta E_p = 17658 \frac{\text{g m}^2}{\text{s}^2} \left(\frac{1 \text{ kg}}{1000 \text{ g}} \cdot \frac{\text{J}}{\text{kg} \cdot \text{m}^2/\text{s}^2} \right) = 17.66 \text{ J}$$

c) increase pressure at 100°C from 0.1 bar to 1 bar

$$\Delta U = m \Delta \hat{U}$$

$$\Delta U = 18 \text{ g} \left(2506.2 \frac{\text{J}}{\text{g}} - 2515.5 \frac{\text{J}}{\text{g}} \right)$$

$$\Delta U = -167.4 \text{ J}$$

d) increase temp from 0°C to 100°C at 5 bar

$$\Delta U = n C_v \Delta T$$

$$\Delta U = 1 \text{ g mol} \cdot 75.4 \frac{\text{J}}{\text{g mol} \cdot ^\circ\text{C}} \cdot (100^\circ\text{C} - 0^\circ\text{C})$$

$$\Delta U = 7540 \text{ J}$$

e) dissociate H_2 and O_2 at 25°C

$$\Delta U_r = \Delta H_r - RT \Delta n_r$$

$$\Delta U_r = 285.84 \frac{\text{kJ}}{\text{g mol}} \cdot \frac{1000 \text{ J}}{1 \text{ kJ}} - 8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}} \cdot 298 \text{ K} (0.5 \text{ g mol})$$

$$\Delta U_r = 284,601.2 \text{ J}$$

f) $E = mc^2$

$$E = \left[1 \text{ g mol} \cdot \frac{18 \text{ g}}{\text{g mol}} \right] \left[3 \times 10^8 \frac{\text{m}}{\text{s}} \right]^2 \left[\frac{1 \text{ kg}}{1000 \text{ g}} \right] \left[\frac{1}{\text{kg} \cdot \text{m}^2/\text{s}^2} \right]$$

$$E = 1.62 \times 10^{15} \text{ J}$$

25/25